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1  """
2  Example 8.4 SVM Regression
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4  """
5
6  import IPython as IP
7  IP.get_ipython().run_line_magic('reset', '-sf')
8
9  import numpy as np
10 import matplotlib.pyplot as plt
11 import sklearn as sk
12 from sklearn import svm
13
14
15 plt.close('all')
16
17 %% build the data sets
18 np.random.seed(2) # 2 and 6 are pretty good
19 m = 100
20 X = 6 * np.random.rand(m,1) - 3
21 y = 0.5 * X**2 + X + 2 + np.random.randn(m,1)
22 y = y.ravel()
23
24 # plot the data
25 plt.figure()
26 plt.grid(True)
27 plt.plot(X,y,'o')
28 plt.xlabel('x')
29 plt.ylabel('y')
30
31
32 %% SVM regression
33
34 svm_reg = sk.svm.SVR(kernel="rbf", degree=3, C=1, epsilon=0.8, gamma="scale")
35 # Try poly kernel, and different degree, C, and epsilon values
36 svm_reg.fit(X, y)
37 x1 = np.linspace(-3, 3, 100).reshape(100, 1)
38 y_pred = svm_reg.predict(x1)
39
40
41 # plot the SVR model on top of the existing data
42 plt.plot(x1, y_pred, "-", linewidth=2, label=r"$\hat{y}$")
43 plt.plot(x1, y_pred + svm_reg.epsilon, "g--", label='curb')
44 plt.plot(x1, y_pred - svm_reg.epsilon, "g--")
45 plt.scatter(X[svm_reg.support_], y[svm_reg.support_], s=100, marker='o', facecolor='none',
46            edgecolors='gray')
47 plt.legend(loc="upper left")

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